PROJECT REPORT

Simulated Temperature Monitoring using ESP32 and Raspberry Pi (MQTT)

Submitted by:

SUJITHA B

- Install ngrok

- sudo apt update
- sudo apt install ngrok

- Add authentication token

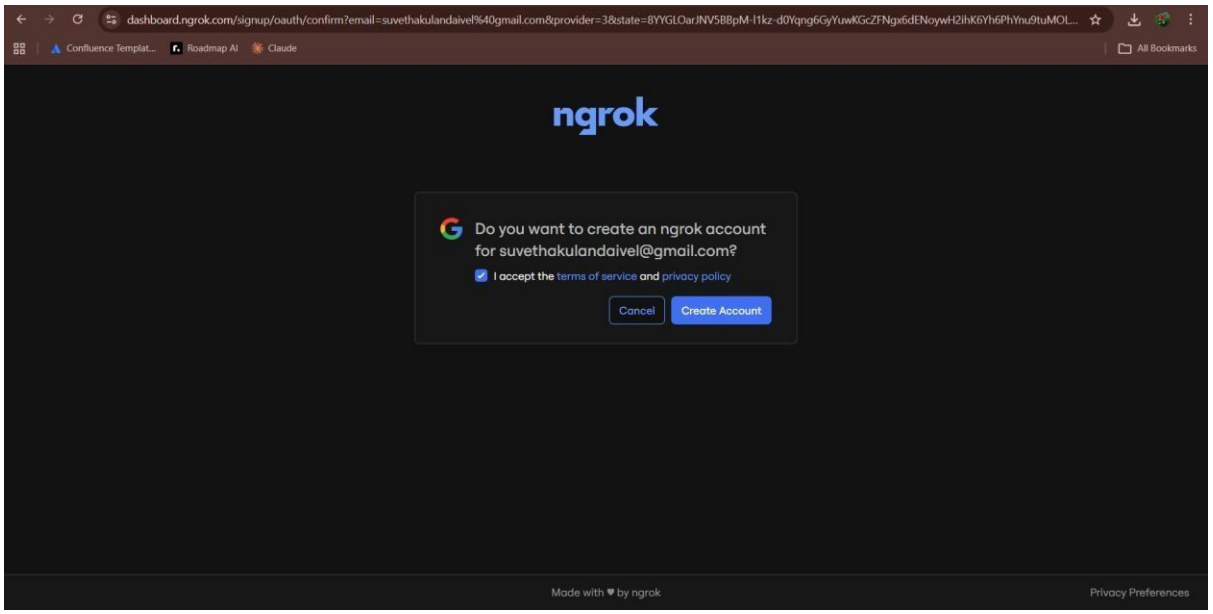
- ngrok config add-authtoken YOUR_TOKEN

- Start web server

- sudo service apache2 start

- Run ngrok

- NgrokHTTP80


dashboard.ngrok.com/signup/oauth/confirm?email=suvethakulandaivel%40gmail.com&provider=3&state=8YYGLOarJNV5BBpM-I1kz-d0Yqng6GyYuwKGcZFNgx6dENoywH2ihK6Yh6PhYnu9tuMOL...
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Do you want to create an ngrok account for suvethakulandaivel@gmail.com?
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Cancel
Create Account
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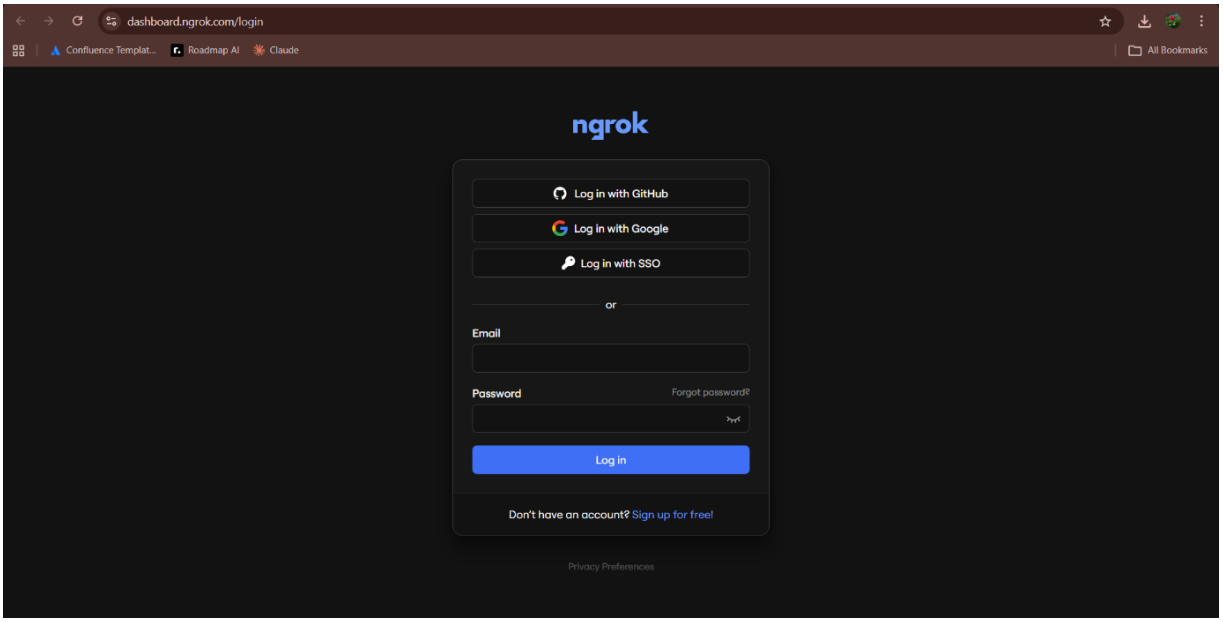
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Log in with GitHub
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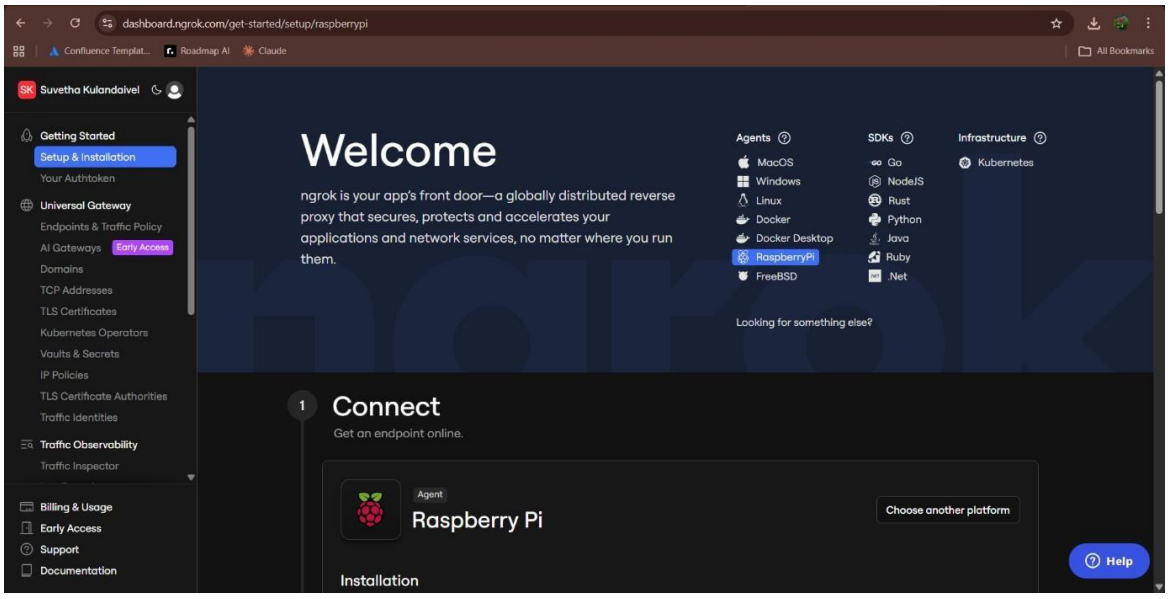
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Getting Started
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Your Authtoken
Universal Gateway
Endpoints & Traffic Policy
AI Gateways
Early Access
Domains
TCP Addresses
TLS Certificates
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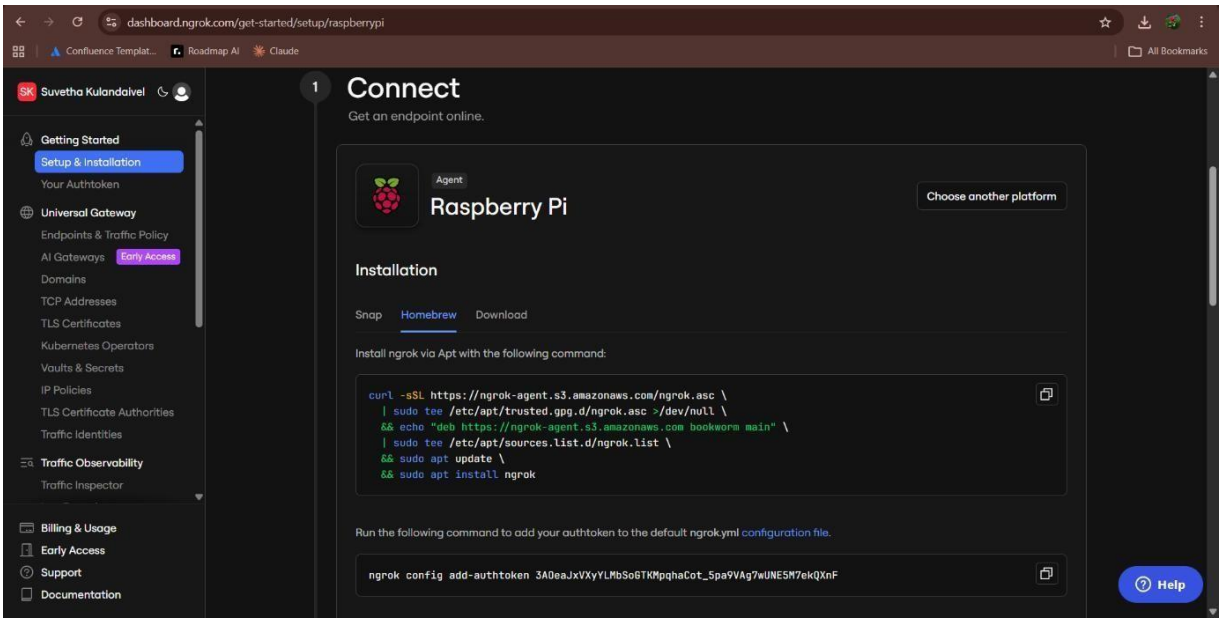
Vaults & Secrets
IP Policies
TLS Certificate Authorities
Traffic Identities
Traffic Observability
Traffic Inspector
Billing & Usage
Early Access
Support
Documentation
Welcome
ngrok is your app's front door—a globally distributed reverse proxy that secures, protects and accelerates your applications and network services, no matter where you run them.
Agents
MacOS
Windows
Linux
Docker
Docker Desktop
RaspberryPi
FreeBSD
SDKs
Go
NodeJS
Rust
Python
Java
Ruby
.Net
Infrastructure
Kubernetes
Looking for something else?
1
Connect
Get an endpoint online.
Agent
Raspberry Pi
Choose another platform
Installation
Help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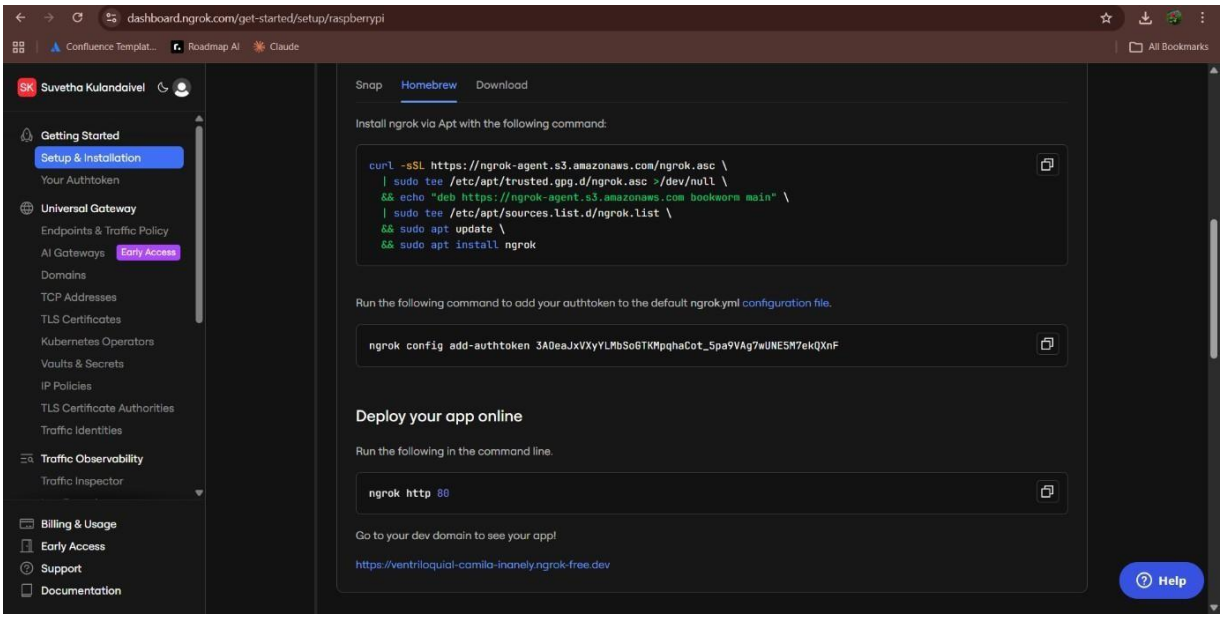
TLS Certificate Authorities
Traffic Identities
Traffic Observability
Traffic Inspector
Billing & Usage
Early Access
Support
Documentation
1
Connect
Get an endpoint online.
Agent
Raspberry Pi
Choose another platform
Installation
Snap
Homebrew
Download
Install ngrok via Apt with the following command:
curl -sSL https://ngrok-agent.s3.amazonaws.com/ngrok.asc \
| sudo tee /etc/apt/trusted.gpg.d/ngrok.asc >/dev/null \
&& echo "deb https://ngrok-agent.s3.amazonaws.com bookworm main" \
| sudo tee /etc/apt/sources.list.d/ngrok.list \
&& sudo apt update \
&& sudo apt install ngrok
Run the following command to add your authtoken to the default ngrok.yml configuration file.
ngrok config add-authtoken 3A0eaJxVXyYLMbSoGTKMpqhaCot_5pa9VAg7wUNE5M7ekQXnF
Help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Early Access
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Download
Install ngrok via Apt with the following command:
curl -sSL https://ngrok-agent.s3.amazonaws.com/ngrok.asc \
| sudo tee /etc/apt/trusted.gpg.d/ngrok.asc >/dev/null \
&& echo "deb https://ngrok-agent.s3.amazonaws.com bookworm main" \
| sudo tee /etc/apt/sources.list.d/ngrok.list \
&& sudo apt update \
&& sudo apt install ngrok
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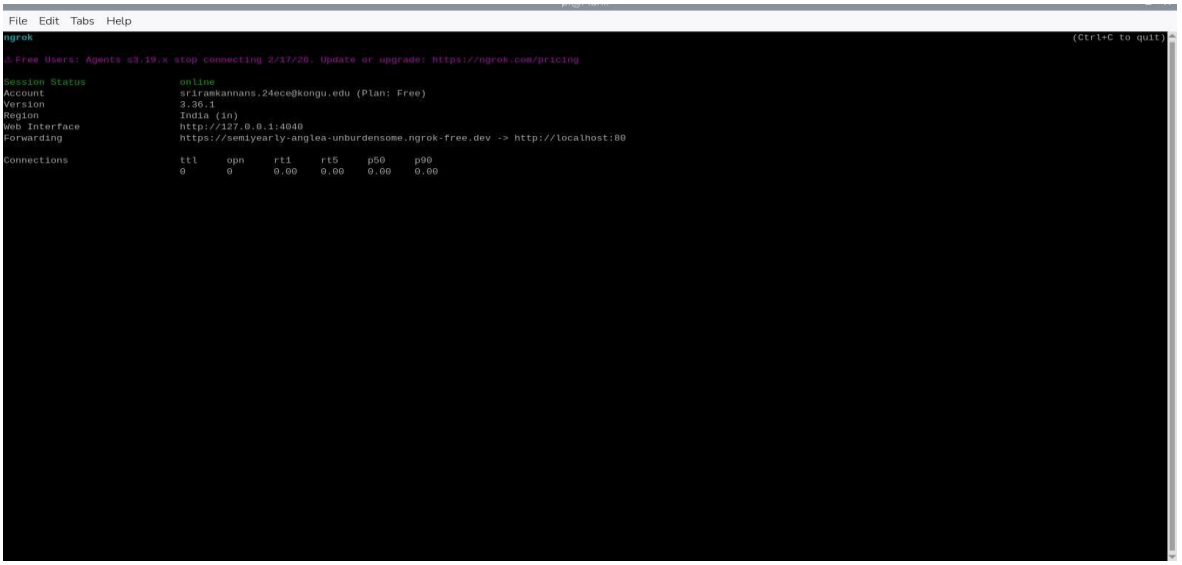
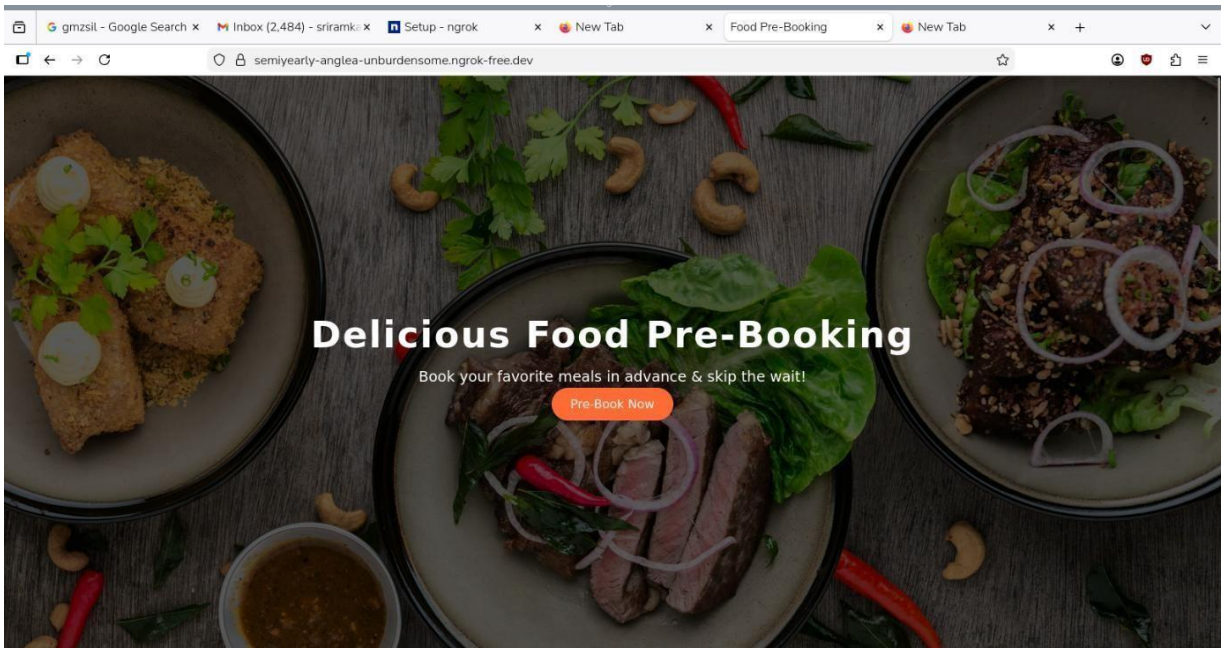
ngrok config add-authtoken 3A0eaJxVXyYLMbSoGTKMpqhaCot_5pa9VAg7wUNE5M7ekQXnF
Deploy your app online
Run the following in the command line.
ngrok http 80
Go to your dev domain to see your app!
https://ventriloquial-camila-inanely.ngrok-free.dev
Help

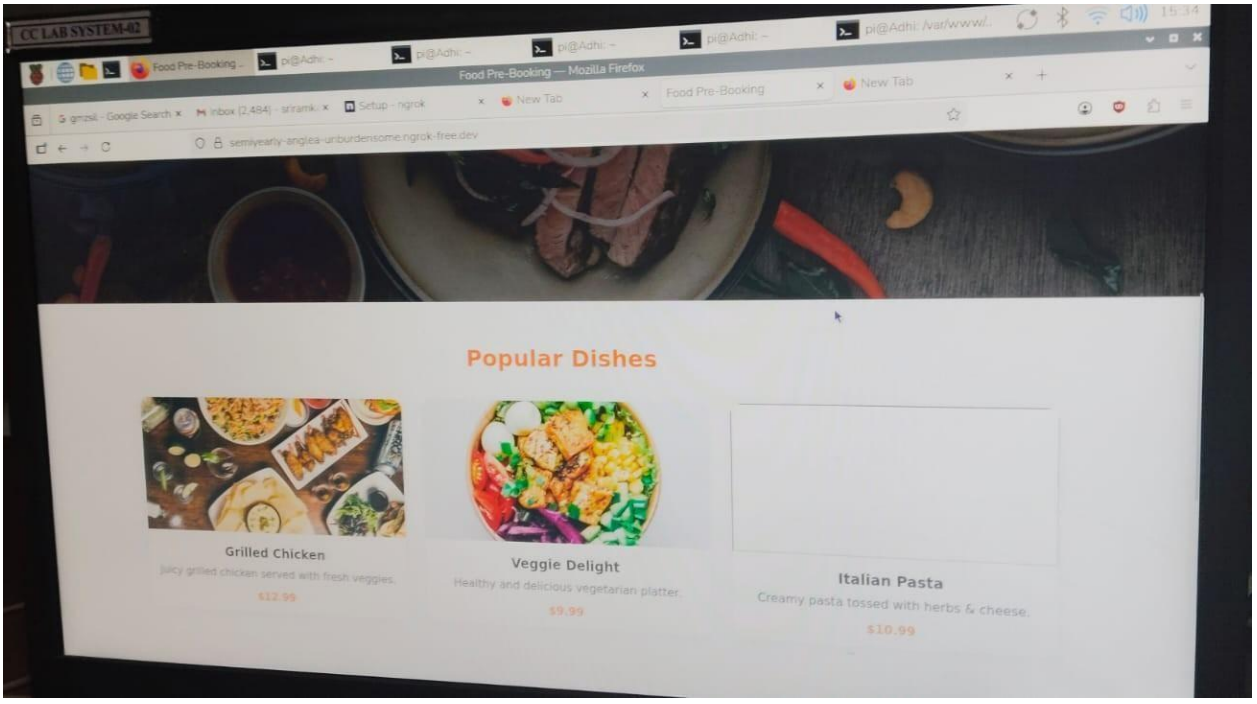

File Edit Tabs Help
ngrok
(Ctrl+C to quit)
Free Users: Agents s3.19.x stop connecting 2/17/26. Update or upgrade: https://ngrok.com/pricing
Session Status online
Account sriramkannans.24ece@kongu.edu (Plan: Free)
Version 3.36.1
Region India (in)
Web Interface http://127.0.0.1:4040
Forwarding https://semiyearly-anglea-unburdensome.ngrok-free.dev -> http://localhost:80
Connections ttl opn rt1 rt5 p50 p90
0 0 0.00 0.00 0.00 0.00

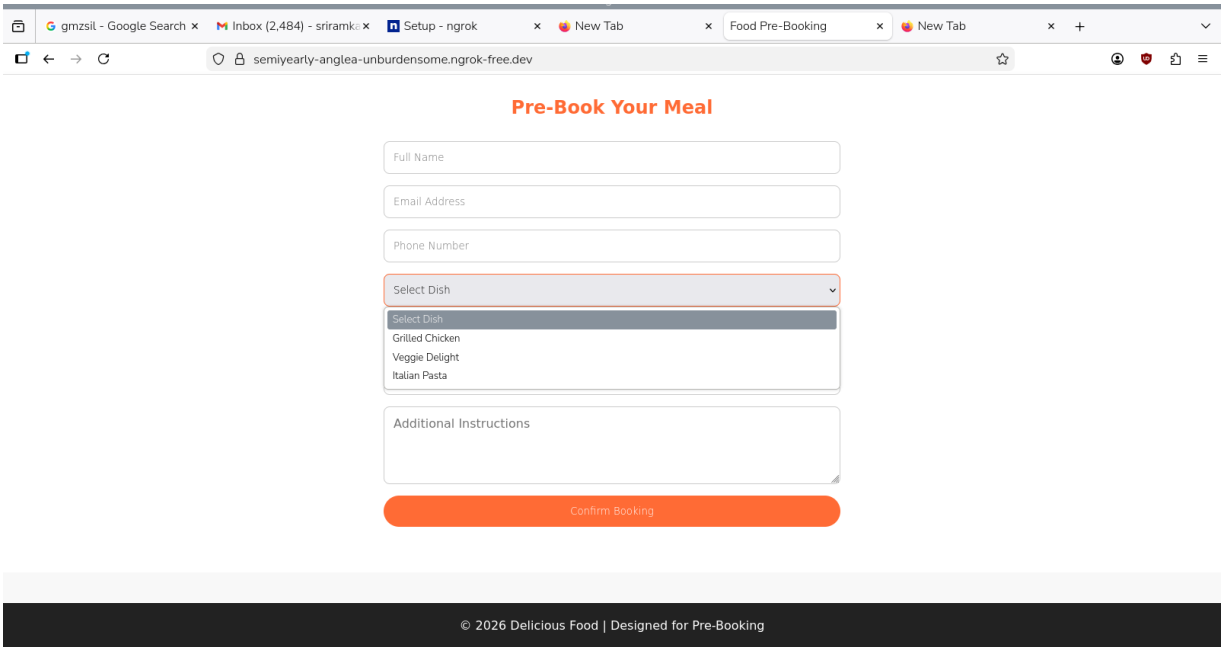
Web Link:

https://semiyearly-anglea-unburdensome.ngrok-free.dev/

Web Output:







Result:

The ESP32 successfully published temperature data and Raspberry Pi received it using MQTT protocol. This demonstrates wireless IoT communication.